

Operations Research

BCA — Semester 4 — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. A milkman buys milk at Rs. 20 per litre and sells at Rs. 25 per litre. Unsold milk has to be thrown away. The daily demand has the following probability distribution: Demand (litres) 46485052545658606264 Probability 0.010.030.060.100.200.250.150.100.050.05 If each day's demand is independent of previous day's demand, using marginal analysis approach, how many litres should be ordered everyday so as to maximize the profit? Also find (a) Max. EMV (b) EPPI (c) EVPI.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Determine the minimum transportation cost from the following per matrix. Stores Warehouse P P P Supply Cost
Unit W 4560453070 W 3515353560 W 3025455590 Demand 60406020220/180

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Solve the given linear programming problem (LPP) by using simplex method and interpret the findings. Min $Z = 20A + 10B$ Subject to $A + 2B \leq 40$ $4A + 3B \geq 60$ $A + B \geq 30$ and $A, B \geq 0$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. There are three jobs P, Q and R to be completed on four machines A, B, C and D. The costs of performing the different jobs on different machines are given below. Assign the jobs to different machines to minimize the total cost of performing the jobs on machines. Jobs Machines A B C D P 90120140160 Q 40658595 R 507595110

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. A TV repairman finds that the time spent on his job has an exponential distribution with mean 30 minutes. If he repairs sets in the order in which they come and if the arrival of sets is approximately Poisson with an average rate of 10 per 8 hour day. What is his expected idle time each day? How many jobs are ahead of the set just brought in?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Assume that two firms are competing for market share for a particular product. Each firm is considering what promotional strategy to employ for the coming period. Assume that the following payoff matrix describe the increase in market share of Firm A and the decrease in market share for Firm B.

Firm A \ Firm B	No promotion	Moderate promotion	Much promotion
No promotion	50-10	Moderate	promotion
Moderate promotion	10-60	Much promotion	20-15
Much promotion	12-20	15-51	12-25

Find the optimal strategy for each firm. Also determine value of the game.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Describe Hungarian Assignment Method (HAM) used for finding the optimal solution of assignment problem.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. The following activities must be completed in order to complete the project. Determine critical path and time duration of the project.

Activity	Predecessor	Time (in week)
A		3
B	A	4
C	A	2
D	B	1
E	C	1
F	D	2
G	E	1
H	F	1
I	G	1
J	H	1

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. A food company at Kathmandu produce three types of a healthy food P, Q and R for divilution which contains three types of vitamin A, B, and C. Each unit of food P contains 2, 2 and 1 unit of vitamin A, B and C. One unit of Q contains 2, 3 and 1 units while each unit of food R contains 1, 1 and 3 units of vitamin A, B and C respectively. Daily minimum requirements of vitamin A, B and C are 10, 12 and 14 units respectively. Formulate objective function and its constraints of LPP if cost per unit of food P, Q and R are Rs. 9, Rs.12 and Rs.15 respectively.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Describe the scope of operations research in different fields.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is called a queue? Describe different queue disciplines.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on: a) Marginal Analysis Approach in decision making. b) Arithmetic Method in game theory.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.